

Matrix Algebra

by Max Kurtz

A *matrix* is a rectangular array of numbers (or letters that represent numbers). The array is enclosed in brackets to indicate its extent, and each matrix is assigned an uppercase boldface letter for identification.

The numbers that compose the matrix are termed its *elements* or *members*. Elements that lie on a horizontal line constitute a *row*, and those that lie on a vertical line constitute a *column*. The rows and columns are assigned identifying numbers by the following convention: number the rows from top to bottom; number the columns from left to right.

The *size* of a matrix is referred to as its *order* or *dimension*, and it is expressed by specifying the number of rows and the number of columns composing the matrix, in that sequence. Thus, a matrix having m rows and n columns is said to be of order $m \times n$, or it is described as an $m \times n$ matrix.

To illustrate the foregoing definitions, consider the following matrix:

$$\mathbf{A} = \begin{bmatrix} 7 & 5 & 6 & 3 \\ -11 & 8 & -3 & 2 \\ 4 & 9 & 0 & -15 \end{bmatrix}$$

This matrix contains 3 rows and 4 columns; therefore, it is a 3×4 matrix. The third row of the matrix is 4, 9, 0, -15 ; the second column is 5, 8, 9.

Where the elements of a matrix are represented by algebraic symbols, the practice is as follows: each element is assigned the lowercase italicized letter corresponding to the matrix label, and two subscripts are appended to specify the location of the element. The first subscript is the row number, the second is the column number. For example, the symbol b_{25} (read " b sub two five") identifies the element in matrix $\mathbf{B}$ that lies in the second row and fifth column. Thus, with reference to the foregoing matrix $\mathbf{A}$, we have $a_{13} = 6$ and $a_{32} = 9$.

A matrix that consists of a single row is called a *row vector*, and one that consists of a single column is called *column vector*. A matrix in which all elements are zero is termed a *null matrix*, and designated as 0.

Consider that we have two sets of items, **A** and **B**. The item in **A** and the item in **B** that have the same identifying number are said to correspond to each other.

Two matrices are *equal* to each other if they are identical in all respects, that is, if they are of the same order and if corresponding elements in the two matrices are equal to each other. If matrices **A** and **B** are equal, we write $\mathbf{A} = \mathbf{B}$.

Consider that one or more rows or columns of a matrix **A** are deleted, and let **B** denote the matrix that remains. Then **B** is a submatrix of **A**. For example, let

$$\mathbf{A} = \begin{bmatrix} -3 & 2 & 9 & -12 \\ 11 & 7 & -4 & 1 \\ 5 & 8 & 6 & 3 \end{bmatrix}$$

If the third row and second and fourth columns of **A** are deleted, the submatrix is

$$\mathbf{B} = \begin{bmatrix} -3 & 9 \\ 11 & -4 \end{bmatrix}$$

A row or column of a matrix is described as *nonzero* if it contains at least one nonzero element.

Consider that a given matrix **A** is transformed by converting its rows to columns and its columns to rows, without disturbing the relative order of the rows and columns. Thus, the *i*th row becomes the *i*th column, and the *j*th column becomes the *j*th row. The matrix that results is known as the transpose of **A**, and it is denoted by $\mathbf{A}^T$ or $\mathbf{A}'$.

For example, if $\mathbf{A} = \begin{bmatrix} 3 & 9 \\ 2 & 6 \\ 4 & 1 \\ 8 & 5 \end{bmatrix}$ then $\mathbf{A}^T = \begin{bmatrix} 3 & 2 & 4 & 8 \\ 9 & 6 & 1 & 5 \end{bmatrix}$

Now consider that a given matrix **A** is transformed by changing the algebraic sign of each element. The matrix that results is the negative of **A**, and it is denoted by $-\mathbf{A}$.

In matrix algebra (as in real-number algebra), parentheses are applied to specify the sequence for performing operations. For example, the notation $(-\mathbf{A})^T$ constitutes the following instruction: form the negative of **A**, and then form the transpose of the negative.

If the transpose of $\mathbf{A}^T$ is formed, the result is the original matrix **A**. Expressed symbolically, $(\mathbf{A}^T)^T = \mathbf{A}$ and therefore, the transpose relationship is reflexive. Similarly, we have $-(-\mathbf{A}) = \mathbf{A}$ and therefore the negative relationship is also reflexive.

Consider that a given matrix $\mathbf{A}$ is transformed to its transpose and the latter is then transformed to its negative. The matrix that evolves is called the *negative transposed matrix*.

There are numerous applications of matrices, both in math and other sciences. For example, 4×4 transformation affine rotation matrices are commonly used in computer graphics. In game theory, the payoff matrix encodes the payoff for two players, depending on which out of a given (finite) set of alternatives the players choose. Early encryption techniques such as the Hill cipher also used matrices. However, due to the linear nature of matrices, these codes are comparatively easy to break.